



MID-TERM EXAMINATION

Course: Introduction to Computer Hardware (23TT-B)

Time: 75 minutes

Term: 3 – Academic year: 2024-2025

Lecturer(s): Dinh Dien

Student name:

Student ID:

(Notes: Neither books nor laptops, phones allowed)

1. (25pts) A logic function is described by the expression:

$$f(x_1, x_2, x_3, x_4) = x_1 \bar{x}_4 + \bar{x}_1 \bar{x}_3 x_4 + \bar{x}_2 \bar{x}_3 \bar{x}_4$$

Implement this function using multiplexer 4:1.

2. (25pts) Design a 4-bit Carry Look-Ahead Adder using basic logic gates(AND, OR, XOR).
3. (25pts) Design a 7-seg decoder using Common Cathode Led7seg for binary code 2, 4, 6, 8, 10, 12 to A, F, d, H, L, P. Don't care the remaining input states. ...
4. (25pts) Using JK-FF and 2-input basic gates(AND, OR) to design a synchronous down counter that count from 8 to 0.

Note: All problems should be explained in your design



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FINAL EXAMINATION
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Course name: Computer Hardware (23TT 1,2) Course code: ECE341
Time: 75 minutes Date: 25/4/2025
Note: Students are [☐ allowed / ☒ not allowed] to use materials during the examination

Full name of Student: Student ID: No:

Question 1 (25 points)

A multiplier circuit employing Booth algorithm is used to multiply two numbers 19 (multiplicand) and -9 (multiplier) expressed in 2's complement binary notation. Show the step-by-step of sequential circuit binary multiplier to calculate the product by using Booth algorithm.

Question 2 (25 points)

Consider three different processors P1, P2 and P3 executing the same instruction set. P1 has 2Ghz clock rate and a CPI of 2.5 . P2 has a 2.5Ghz clock rate and a CPI of 2.2. P3 has a 3.0 Ghz clock rate and has a CPI of 3.2

- Which processor has the highest performance expressed in instructions per second?
- If the processors each execute a program in 24 seconds, find the number of cycles and the number of instructions.
- We are trying to reduce the execution time by 30% but this leads to an increase of 20% in the CPI. What clock rate should we have to get this time reduction?

Question 3 (25 points)

Design a Moore FSM to detect serial bit stream input. The FSM should output $z=10$ when the input pattern 1101 is detected, and $z=01$ when the input pattern 1011 is detected ,including overlap patterns. In all other cases, the output should be $z=11$.

Question 4 (25 points)

The 5-stage processor is used to execute the following sequence of instruction, one after other:

Instruction 1: Load R5, 400(R4)

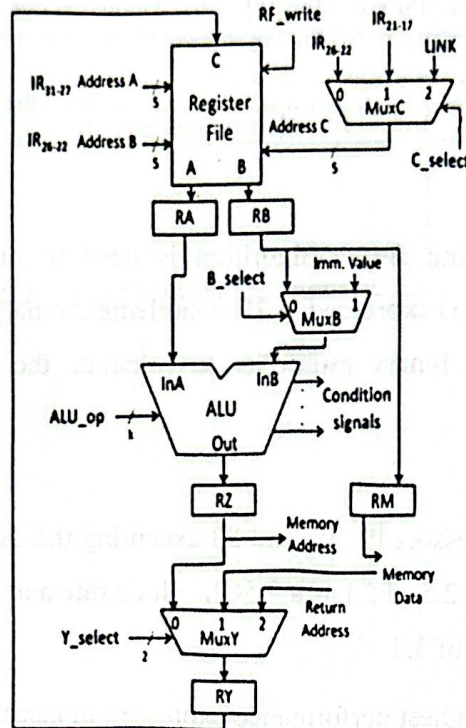
Instruction 2: Xor R6, R3, R5

Instruction 3: Store R6, 600(R4)

The processor datapath with all control signals is shown in following figure:

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Assume that the final content of registers R3, R4, R5 and R6 are 200, 28000, 800 and 70 respectively. The contents of memory locations at address 28000, 28200, 28400 and 28600 are 400, 500, 600 and 700 respectively.

- a. For each of three instructions in the program, answer the following questions:
- Which of two inputs is selected by **MuxB**?
 - What are the contents of register **RY** after complete stage-4 (memory access stage)?
 - Write down the value of control signal **RF_write** during stage-5 of instruction processing.
- b. After the completion of the above code sequence, what the final contents of memory locations at addresses 28400 and 28600?

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